Extended reading — Week 6: Rainbow DQN and BBF

STUDY-RL

November 8, 2026

Read with a question in mind

Start with the week's teaching guide and its small worked example. Then use these readings to connect the algorithm to its assumptions and evidence. The explanations and investigations below are course commentary, not quotations. Read the source before reusing a theorem or a reported result. For each paper, record the objective, observation model, claimed result, conditions and one condition your local experiment does not reproduce.

1. Hessel et al. (2018): Rainbow

Primary source or book record: https://arxiv.org/abs/1710.02298.

What to learn. Combines six developments in deep Q-learning and evaluates their contributions through ablations.

What the result requires. An ablation effect depends on the remaining components, environment and budget. A local teaching implementation can include a subset or simplified variant without reproducing published scores.

Notebook investigation. Use `exercise_9_smoke_ablate`. Confirm that each switch actually changes the computation before attributing a score difference to that component.

2. van Hasselt, Guez and Silver (2016): Deep Reinforcement Learning with Double Q-learning

Primary source or book record: https://arxiv.org/abs/1509.06461.

What to learn. Uses the online network for next-action selection and a target network for evaluation.

What the result requires. Correlated errors remain possible, and the construction is not a guarantee of an unbiased target. A double-network method is not automatically the minimum-of-two rule.

Notebook investigation. Use `exercise_1_double_dqn_target`. For online values $(5, 4)$ and target values $(1, 3)$, the selected next value is 1; an ordinary target maximum would be 3.

3. Schaul et al. (2016): Prioritized Experience Replay

Primary source or book record: https://arxiv.org/abs/1511.05952.

What to learn. Samples replay entries nonuniformly and uses importance weights to control the resulting change in the replay objective.

What the result requires. At full correction, weights undo replay-sampling bias toward high-priority entries. They do not correct the action probabilities of the behaviour policy or make a bootstrapped target unbiased.

Notebook investigation. Use `exercise_3_per_weight` and `exercise_4_priority_update`. Keep sampling probability $P(i)$ distinct from a policy propensity $\mu(a \mid s)$.

4. Bellemare, Dabney and Munos (2017): A Distributional Perspective on Reinforcement Learning

Primary source or book record: https://arxiv.org/abs/1707.06887.

What to learn. Learns a return distribution and introduces categorical distributional control with projection onto a fixed support.

What the result requires. Distributional policy-evaluation contraction and control behaviour need separate statements. Projection can lose tail information outside the support, and the mean of a distribution is not a risk guarantee.

Notebook investigation. Use `exercise_6_categorical_project`. Verify total mass remains one, including a target exactly on an atom. A terminal target collapses continuation before projection.

5. Schwarzer et al. (2023): Bigger, Better, Faster

Primary source or book record: https://proceedings.mlr.press/v202/schwarzer23a.html.

What to learn. Investigates scaling value networks with a coordinated recipe for sample-efficient Atari learning, including replay, representation learning and resets.

What the result requires. The published result is a benchmark evaluation, not a 2025 universal frontier or evidence that adding two switches reproduces BBF. Sample efficiency and wall-clock efficiency require different denominators.

Notebook investigation. Use `exercise_8_shrink_perturb` to inspect a reset in isolation. Report parameter change and the role of fresh noise; distinguish this toy mechanism from the full published training system.

Synthesis: evidence you should be able to show

1. Reproduce one numerical calculation before running a long experiment.
2. Name the precise objective, sampling distribution and terminal convention.
3. Compare the local implementation with one source: list a shared mechanism and one simplification that prevents calling it a full reproduction.
4. Separate an algebraic identity, an assumption-dependent theorem and an empirical result. A plot alone does not prove a theorem.
5. Save the notebook with seed, parameters, axes, denominators and a short explanation of an unexpected result. Keep evaluation data separate from tuning.

Return to the teaching guide's notebook investigation for the local function names, expected checks and interpretation questions. Source code and notebooks are available together in the course repository; the web reader is the entry point and the downloaded notebook is the execution workspace.

Statistics reading bridge

A return distribution is not an uncertainty interval for its mean. Read outcome variance and variance of a sample mean as two distinct distributions. Relate that distinction to distributional targets; use the primary C51/IQN sources for their losses and approximation results.

Checked textbook locations: Siegrist § § 4.3, 6.2, PDF pp.460, 817. https://stats.libretexts.org/@go/page/10158. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.